$G_t$ (matrix gradient)

$R \leftarrow \beta_2 R + (1-\beta_2)(G_t^2)1_n^T$

$C \leftarrow \beta_2 C + (1-\beta_2)1_m^T(G_t^2)$

$\hat{V} = (R \cdot C)/(1_n^T R)$

$U_t = G_t/(\sqrt{\hat{V}} + \varepsilon)$

$\hat{U}_t = U_t/\max(1, \mathrm{RMS}(U_t))$

$\theta_{t+1} \leftarrow \theta_t - \eta\hat{U}_t$